

(c) Solve by method of variation of parameters given that $y'(0) = 0$, $y''(0) = 6$.

IET, DAVV, Indore

B.E. II (Civil)

Class Test I (August-2023)

3AVRCI: Applied Mathematics-III

Time: 70 min

Note: Attempt any **FOUR** questions. All questions carry equal marks.

Max Marks: 20

Q.1. Evaluate

$$\text{a) } L \left\{ t \int_0^t e^{-t} \frac{\sin t}{t} dt \right\} \quad \text{b) } \int_0^\infty e^{-t} \left(\frac{\cos at - \cos bt}{t} \right) dt \quad (05)$$

Q.2. Using convolution theorem to find:

$$\text{a) } L^{-1} \left\{ \frac{1}{s^2(s+1)^2} \right\} \quad \text{b) } L^{-1} \left\{ \frac{s^2}{(s+4)^2} \right\} \quad (05)$$

Q.3. Using the method Laplace transform, solve the equation: $(D^2 - 3D + 2)y = 1 - e^{2t}$ (05)

Q.4. Obtain a root of equation $\cos x = xe^x$ using **Bisection method**, correct to three decimal places. (05)

Q.5. Find a real root of the equation $x \log_{10} x = 1.2$ by **Regula falsi method**, correct to four decimal places. (05)
